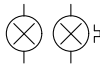
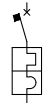
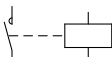
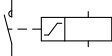


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TENDRARA PROJECT GAS TREATMENT AND LNG PRODUCTION PLANT ON RENTAL BASIS																					
SAFETY CRITICAL ELEMENT LIST																					
IFI	2	27/02/2026	AS-BUILT				ADF	GDP	IC												
IFI	1	24/03/2025	GENEERAL REVIEW				ADF	GDP	IC												
IFI	0	26/03/2024	ISSUED FOR COMMENTs				ADF	GDP	IC												
Status	Rev. n.	Rev. n.	Rev. n.				Rev. n.	Rev. n.	Rev. n.												
Rev. Index		Date	Description				Prepared	Checked	Approved												
Client							Project Title														
ManaEnergy Managem Group MANAENERGY Mahaj Riad Center, Batiment 6, 3eme etage Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco							TENDRARA PROJECT														
Contractor							Provision of a m-LNG plant for production for ManaEnergy														
ITALFLUID ITALFLUID GEOENERGY Marocco Branch Boulevard de Bourgogne Rue Jaafar Ibn Habib Residence Al Mashrik II 1ere Etage N3 Casablanca																					
Engineering																					
ITALFLUID ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) Tel:085-4224374 / fax: 085-4220742																					
Facility Name			Location				Scale		Sheet of Sheets												
TENDRARA			MOROCCO				A4		1 of 13												
Document Title					Doc. Code																
P-PN-0S331.1					ELD01-ING-FR01																
AMMINE GAS -GAS HEAT EXCHANGER					Vendor Doc. code																
					IE24090_SE27_P-PN-OS331.1																
			Project : P–PN–0S331.1				Customer : ManaEnergy Managem Group				Object :										
General Contractor :		Manufacture : esa esaelettromeccanica.com		AMMINE GAS –GAS HEAT EXCHANGER				Contract : IE24090				Rev.		Data : 27/02/2026				Date : 26/03/2024			
								S/N : IE24090_SE27				Rev. 1		Data : 24/03/2025				Sheet : 1		of : 13	

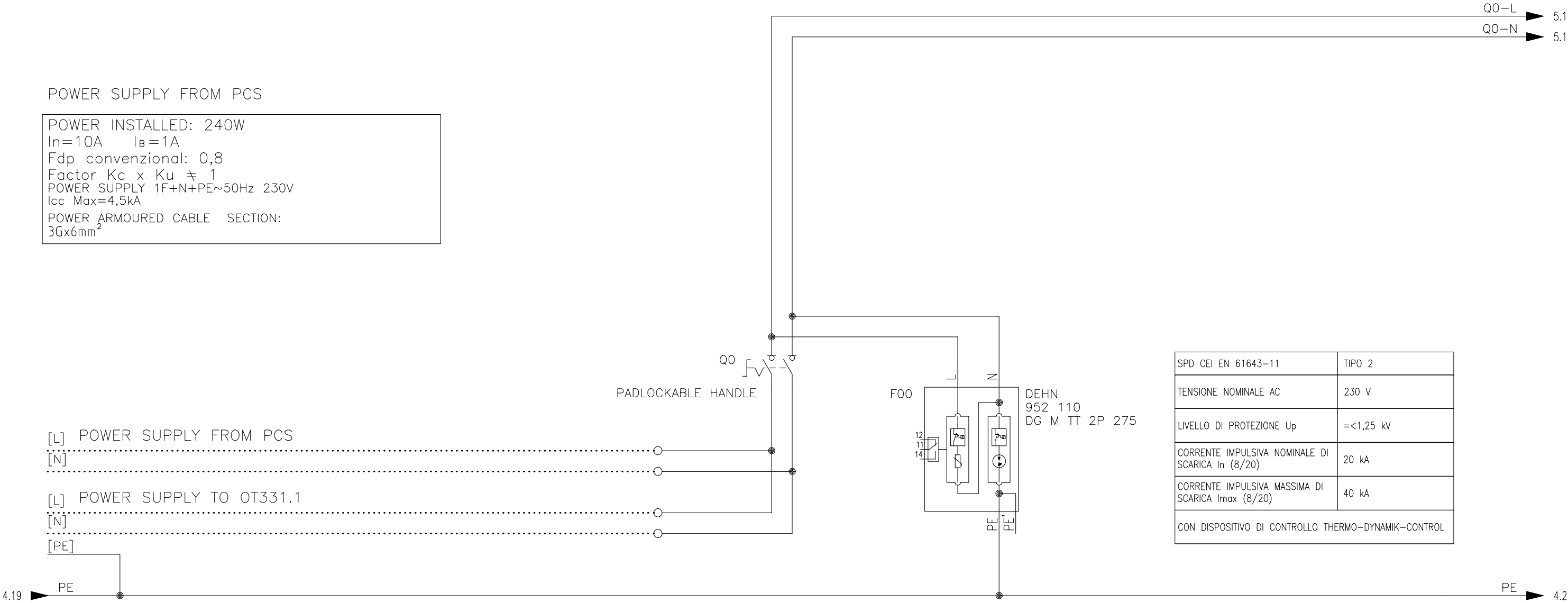
CIRCUIT SYMBOLS ACCORDING TO CEI STANDARD

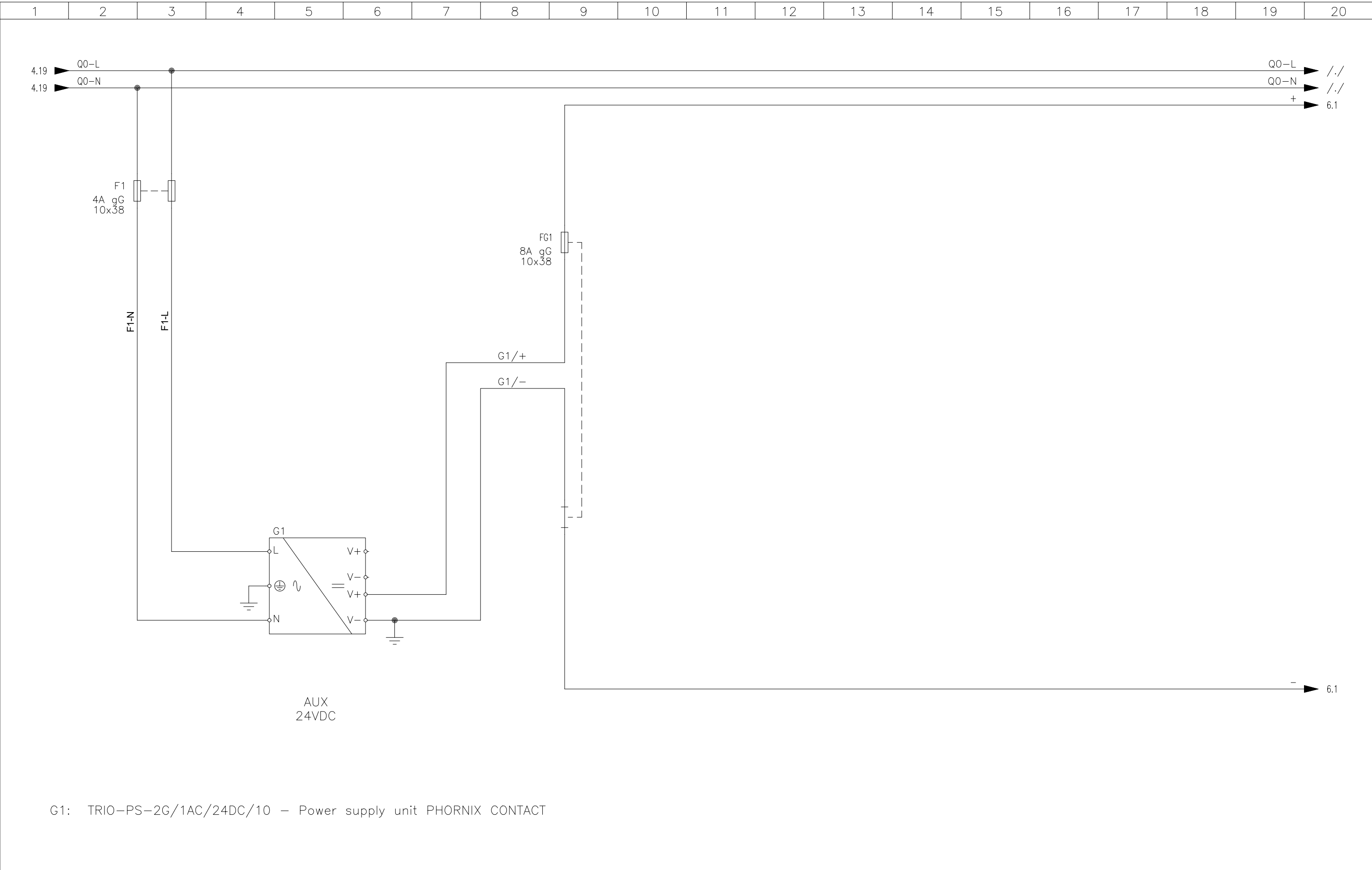
	ISOLATOR		ELECTRONIC TIMER WITH PICK-UP DELAY		MUSHROOM PUSH-BUTTON PULSE CONTACT WITH STAY-PUT MAKE CONTACT OPERATED BY STRIKING		STEDY OR FLASHING LIGHT
	FUSED ISOLATOR		ELECTRIC PULSE METER		KEY OPERATING DEVICE FOR SELECTORS, PUSH-BUTTON, ETC.		PLC UNIT
	CIRCUIT BREAKER WITH THERMAL AED MAGNETIC OVERCURRENT RELEASES		FUSE CARRIER		PRESSURE SWITCH		CONTROL INSTRUMENT
	CIRCUIT BREAKER WITH MAGNETIC RELEASE		MAKE CONTACT OF CONTACTOR		TEMPERATURE SWITCH		ENERGY METER
	CIRCUIT BREAKER WITH THERMAL, MAGNETIC AND DIFFERENTIAL RELEASES		MAKE CONTACT OF CONTACTOR		FLOW SWITCH		MEASURING INSTRUMENT
	RESIDUAL CURRENT CIRCUIT BREAKER "AC" TYPE FOR ALTERNATING CURRENT "A" TYPE FOR ALTERNATING OR PULSATING CURRENTS		MAKE AND BREAK CONTACT		FLOAT SWITCH		VOLTAGE TRANSFORMER
	THERMAL RELAY (OVERLOAD RELAY)		MAKE AND BREAK CONTACT WITH DROP-OUT DELAY		BUCHHOLZ RELAY		CURRENT TRANSFORMER
	COIL OF CONTACTOR RELAY		MAKE AND BREAK CONTACT WITH PICK-UP DELAY		CAM OPERATED DEVICE		ACOUSTIC SIGNALLING DEVICE
	COIL OF SHUT-OFF ALARM MAKE CONTACT ON ENERGIZATION		MAKE AND BREAK CONTACT WITH DROP-OUT AND PICK-UP DELAY (FLIP-FLOP)		PROXIMITY SWITCH		RESISTANCE
	COIL OF SHUT-OFF ALARM DENERGIZES IN CASE OF UNDERVOLTAGE		MAKE AND BREAK CONTACT OF THERMAL RELAY		TEMPERATURE SENSOR		CAPACITOR
	COIL OF OPERATING DEVICE WITH DROP-OUT DELAY		MAKE AND BREAK CONTACT OF LIMIT SWITCH		PHOTOELECTRIC SENSOR		SINGLE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH PICK-UP DELAY		MULTI POSITION SWITCH (GENERAL SYMBOL)		SOLENOID VALVE		THREE-PHASE ELECTRIC MOTOR
	COIL OF OPERATING DEVICE WITH DROP-OUT AND PICK-UP DELAYS		MAKE AND BREAK CONTACT OPERATED BY TURNING WITH SPRING RETURN		PHOTOCELL		
	COIL OF LATCHING RELAY (STEP-BY-STEP FUNCTION)		MAKE AND BREAK CONTACT OPERATED BY PUSHING WITH SPRING RETURN				
	ELECTRONIC OR ELECTRO-MECHANICAL TIME SWITCH		MAKE AND BREAK CONTACT OPERATED BY PULLING WITH OR WITHUOT SPRING RETURN				

CONDUCTOR COLOR CODING INSIDE CONTROL PANEL	
AC AND DC POWER CIRCUIT	BLACK
NEUTRAL CONDUCTOR OF POWER CIRCUIT	LIGHT-BLUE
PROTECTIVE CONDUCTOR (EARTH CONDUCTOR)	YELLOW-GREEN
110/220V AC AUXILIARY CIRCUIT (*)	RED
24V AC AUXILIARY CIRCUIT	WHITE
24V AC AND DC PLC INPUT	GRAY
24V AC PLC OUTPUT	WHITE
24V AC AND DC INTERLOCKING CIRCUIT	ORANGE
24V DC AUXILIARY CIRCUIT AND PLC OUTPUT	DARK-BLUE
(*) : RED FOR 110V AND BROWN FOR 220V, IN CASE BOTH VOLTAGE ARE USED	

CODE COLOUR	
COLOUR	CODE LETTERS
BLACK	BK
BROWN	BN
RED	RD
ORANGE	OG
YELLOW	YE
GREEN	GN
BLUE	BU
GREY	GY
WHITE	WH
SILVER	SR
CLEAR	CL
GREEN AND YELLOW	GNYE

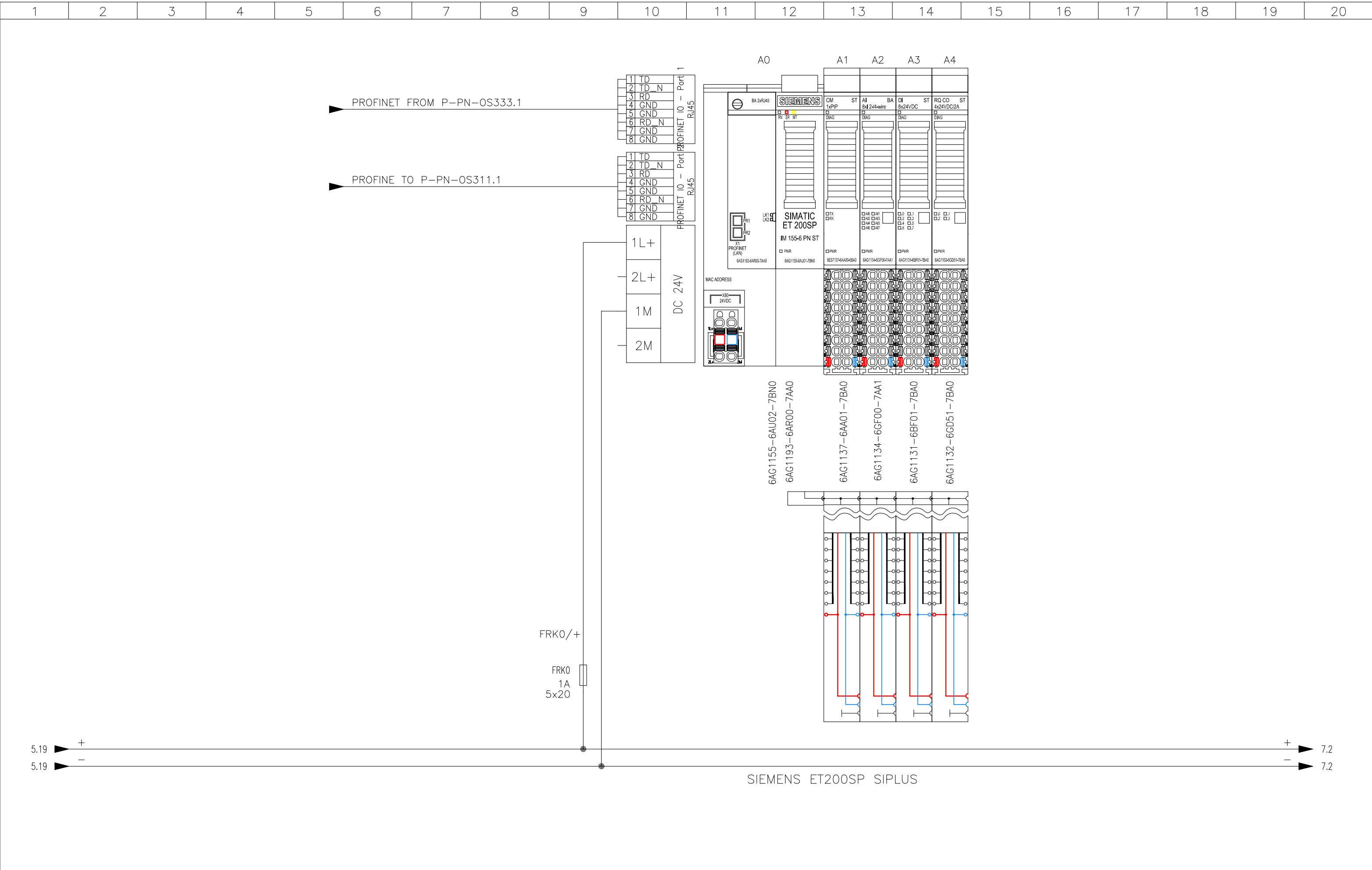
	POWER MAIN DISCONNECT SWITCH																FEEDER ID		
	1F+N+PE~50Hz 230V																RATED POWER Pn~RATED VOLTAGE Vn		
																	RATED CURRENT		





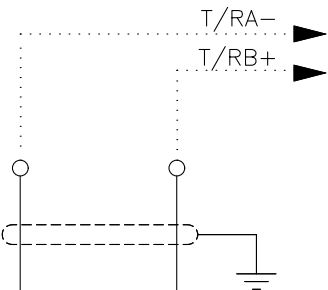
G1: TRIO-PS-2G/1AC/24DC/10 - Power supply unit PHORNIX CONTACT

General Contractor :		Manufacture : 	Project : P-PN-0S331.1 AMMINE GAS –GAS HEAT EXCHANGER	Customer :  Contract : IE24090 S/N : IE24090_SE27	Object :			
					Rev.	Data : 27/02/2026	By :	
					Rev.	Data : 24/03/2025	Date : 26/03/2024	
					Rev.	1	Data : 24/03/2025	Sheet : 5 of : 13

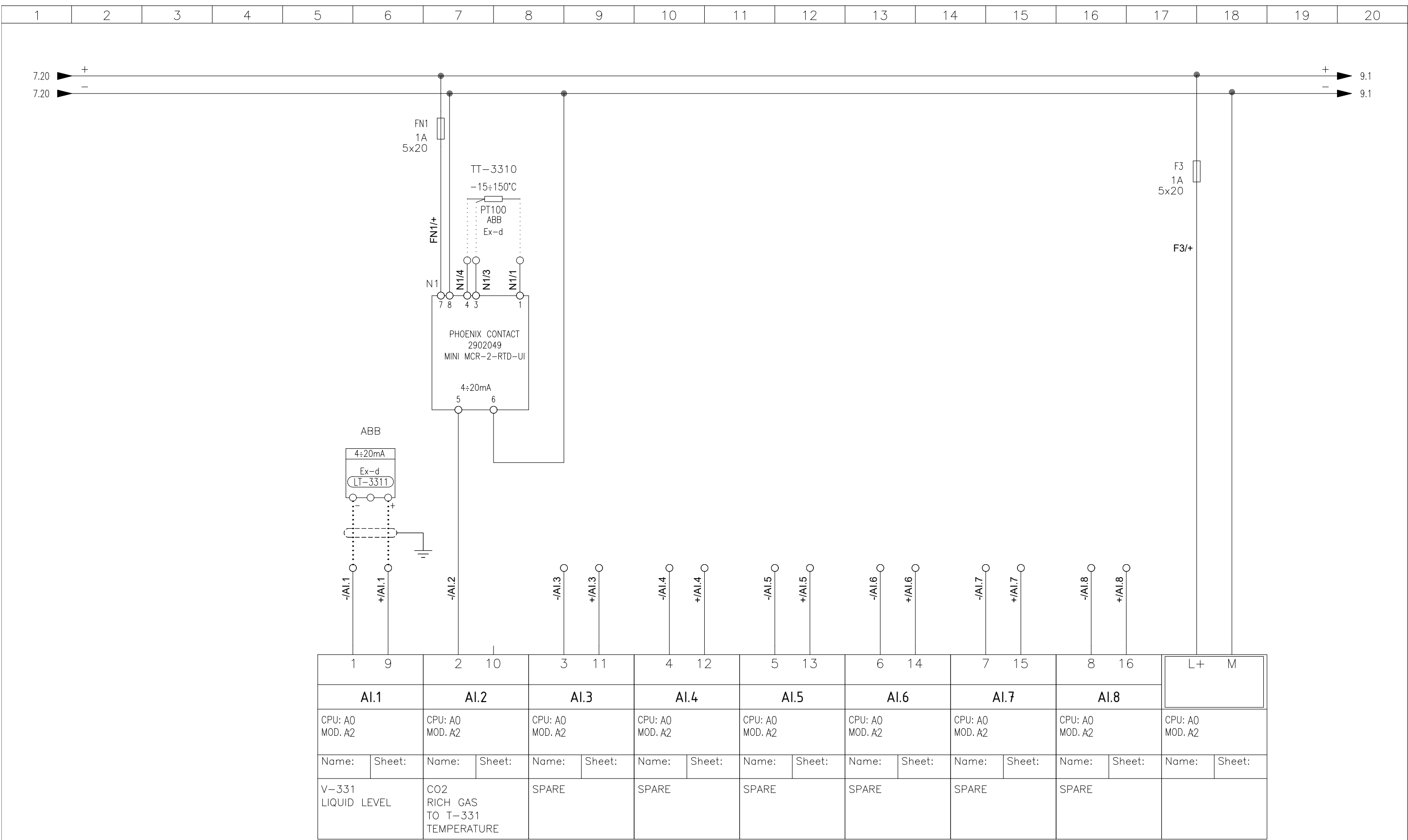


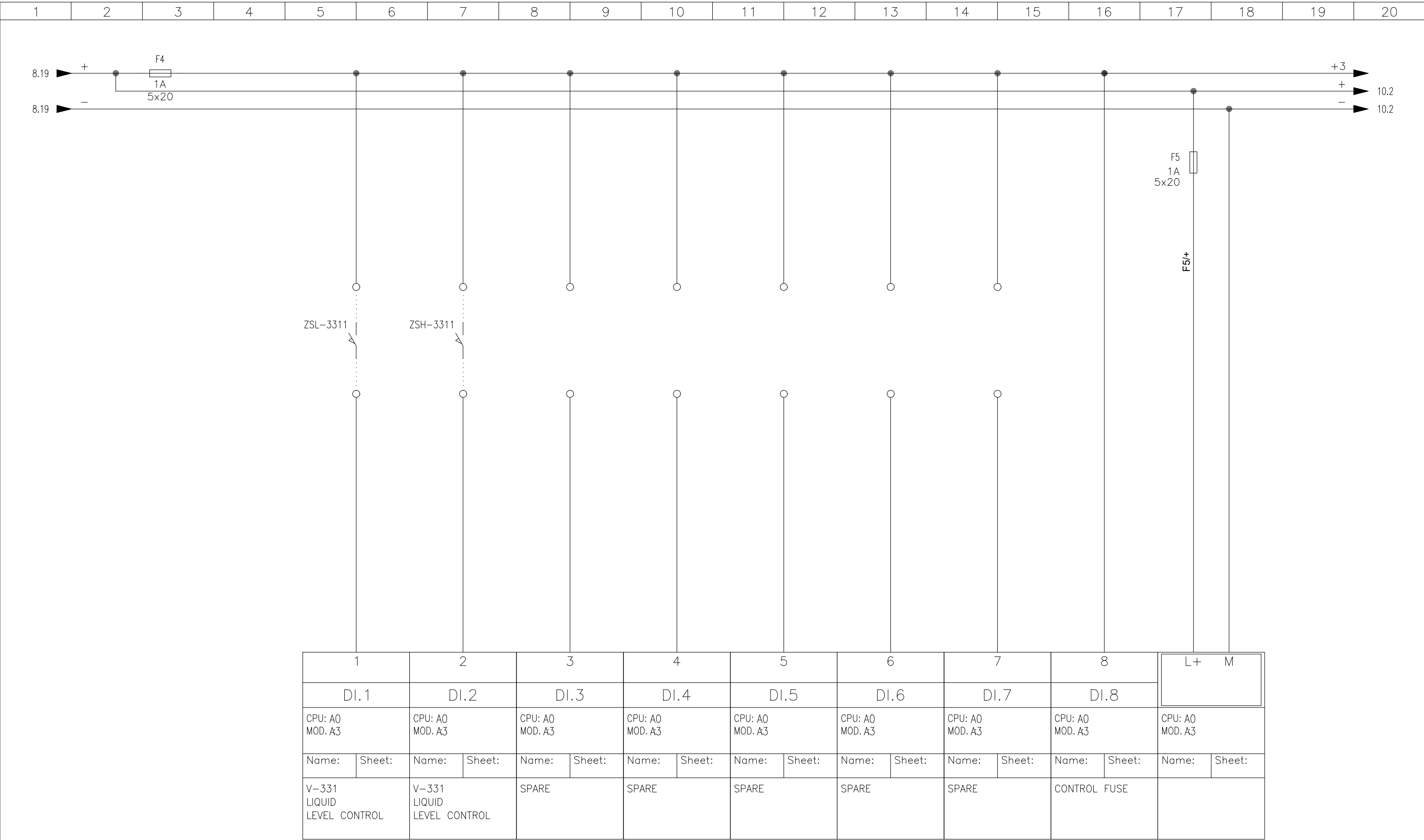


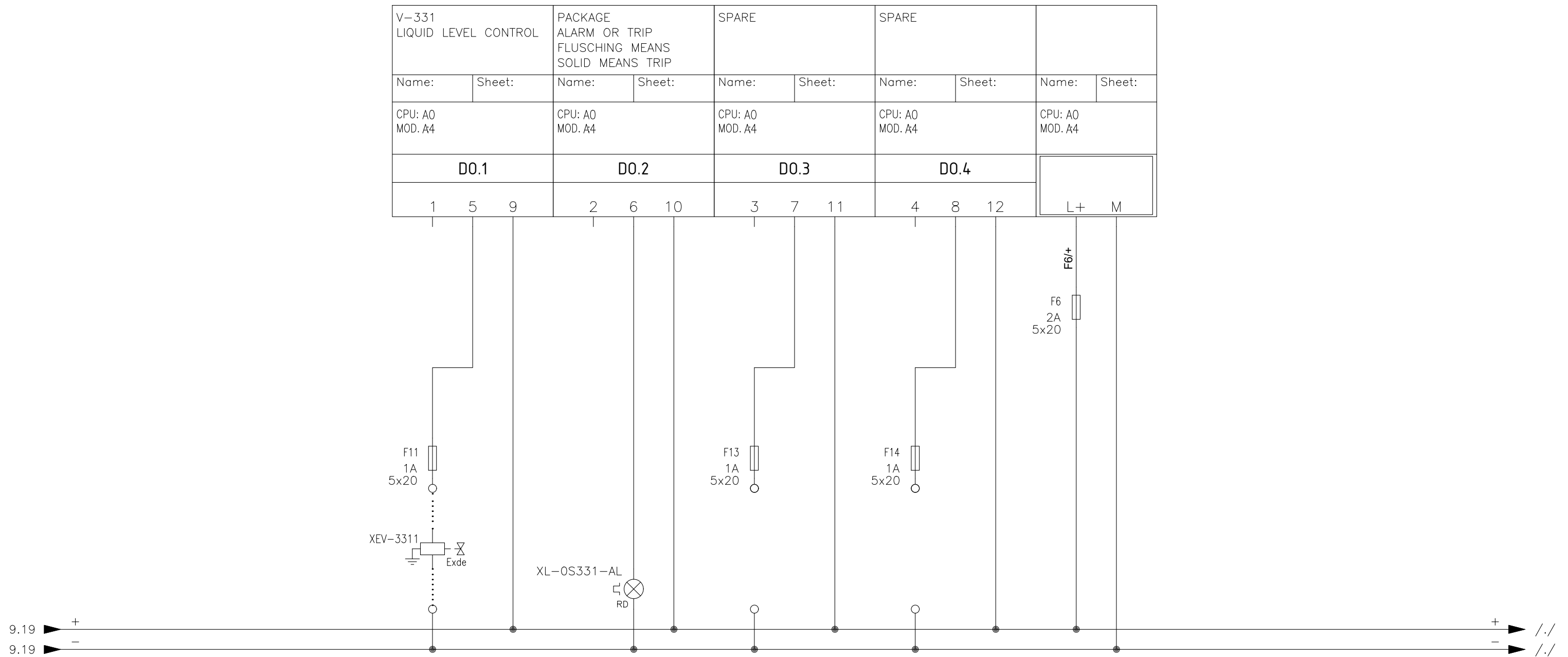
Name:					Sheet:				
CPU: A0 MOD. A1									
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11	12	13	14	15	16	L+	M
CPU: A0 MOD. A1						CPU: A0 MOD. A1	
Name:			Sheet:			Name:	Sheet:
SWEETENING UNIT OUTLET CO2 CONCENTRATION							







1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20									
INITIALS	SUPPLIER PN		DESCRIPTION															SUPPLIER										
–	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT										
+	3273110		PTFIX 6/18X2,5–NS35 GY – Distribution block															PHOENIX CONTACT										
A0	6AG1155–6AU01–7BN0		ET 200SP, PROFINET interface module IM 155–6PN Standard															SIEMENS										
A0	6AG1193–6AR00–7AA0		ET 200SP BA 2xRJ45															SIEMENS										
A1	6AG1137–6AA01–7BA0		ET 200SP CM PTP															SIEMENS										
A1	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals															SIEMENS										
A2	6AG1134–6GF00–7AA1		AI 8xl 2–/4–wire BA															SIEMENS										
A2	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals															SIEMENS										
A3	6AG1131–6BH01–7BA0		ET 200SP DI 16x24 V DC standard															SIEMENS										
A3	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals															SIEMENS										
A4	6AG1132–6GD51–7BA0		ET 200SP RQ 4x24VDC/2A CO ST															SIEMENS										
A4	6AG1193–6BP00–7DA0		ET 200SP BU15–P16+A0+2D push–in terminals															SIEMENS										
F00	952110		DG M TT 2P 275 – SPD															DEHN										
F1	2302038		Fuseholder for cylindrical fuses – 2P for fuses 10,3x38mm – BCH 2x38 32A 690V															ITALWEBER										
F11	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F13	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F14	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F2	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F3	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F4	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F5	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
F6	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
FG1	2301138		Fuseholder for cylindrical fuses – 1P+N for fuses 10,3x38mm – BCH 1x38N 32A 690V															ITALWEBER										
FN1	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
FRK0	TBS4F		MODULAR TERMINAL WITH FUSE															CEMBRE										
G1	2903149		TRIO–PS–2G/1AC/24DC/10 – Power supply unit															PHOENIX CONTACT										
N1	2902049		MINI MCR–2–RTD–UI – Resistance thermometer measuring transducer															PHOENIX CONTACT										
P–PN–0S331.1	EJB–E		ENCLOSURE 540x594x311 mm L x H x D															CMC										
Q0	PSRC–320		SELECTOR SWITCH															CMC										
XL–0S331–AL	PLD20		PILOT LAMP RED															CMC										
XL–0S331–AL	O20PD9		LAMPHOLDER BA9S, DIRECT SUPPLY															ELFI										
General Contractor : Manufacture : 			Project : P–PN–0S331.1 AMMINE GAS –GAS HEAT EXCHANGER					Customer :  Contract : IE24090 S/N : IE24090_SE27					Object :				By :											
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